

CHAT

COMPUTATIONALLY HUMANISED AUDIO TO TEXT



ALEXANDER EINODER -S3866647

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Ali Einoder
s3866647

INTRODUCTION

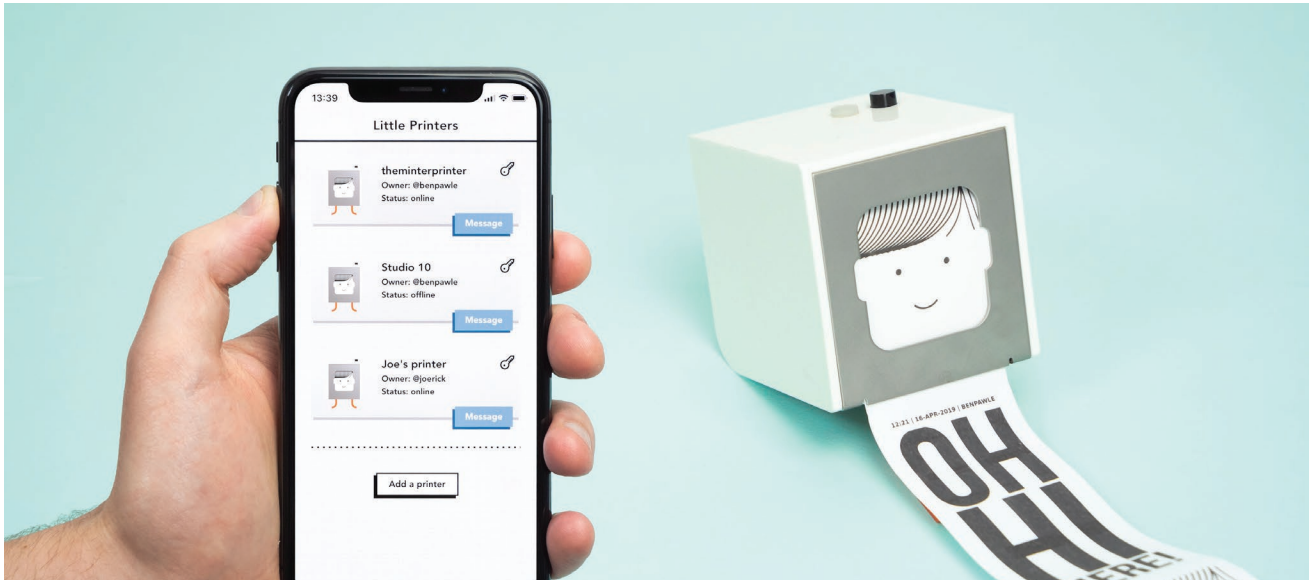
The ambiguity and detached nature of written text has become a burden of the modern era. Constant conflict is created both directly and indirectly by a lack of many human qualities in text. The inability to assess the non-verbal qualities of the communication leaves the reader without the true sense of the emotional, human qualities of the author (James 2017). This in turn can lead the reader to respond and act in a manner which may contradict the real-world, face-to-face reaction they would have if witnessing the author speak this text.

Through computationally processing the spoken word, the CHAT team aims to provide insight into potential ways written text can evolve to become a truer representation of spoken language.

How does the medium in which the text is viewed affect the reader's emotional connection to the speech? If the text is formed into a physical artifact will the reader attribute different emotional qualities to the text? How can the typography be altered to represent the nonverbal aspects of human speech?

These questions and more were explored through research and experimentations over the course of the semester, resulting in the development of CHAT - a device which computationally humanises typography through audio to text translation. Using technologies such as speech to text, electromagnetic field recording and audio analysis, the team mapped non-verbal language to the textual representation of the verbal. This was then built into a zoomorphically designed object that prints these messages (along with an image of the sender) on thermal paper to create a timeline of conversation that now occupies the physical world as a receipt of the interaction.

We hope that this device illustrates the potential for further development in the text-based world, especially given the move to a greater online presence.



PRECEDENTS

LITTLE PRINTER BERG 2012

Little Printer was a small thermal printer that was connected to the internet and able to print out web articles, social content, and certain images. This printer used Berg's cloud infrastructure to organize content and format it to be printed from the thermal printer (Nord Projects n.d.).

After Berg closed their servers, this project was resuscitated by Nord Projects and was given the ability to connect to many people who can send content directly to a printer. It was also given the ability to print images sent from a phone after an Atkinson Dithering process was applied to allow for 1bit printing.

PROSODIC FONT TARA MICHELLE GRABER ROSENBERGER 1998

This thesis details the issues with written text in comparison to the spoken word. Specifically focussing on the prosody of language and its lack of representation in writing. It aims to create a method of converting the spoken word into a moving font that illustrates how a speaker said the words that are being presented (Rosenberger & MacNeil 1999).

By using computationally derived values from speech analysis, the author dictated the creation of a new font that is systematically constructed, letter by letter using a variation in four key components of the letters.

ETCH-A-SKETCH THEREMIN LOOK MUM NO COMPUTER 2019

The youtuber/inventor Sam Battle, under his moniker LOOK MUM NO COMPUTER, used an amalgamation of vintage audio test equipment, a Moog Etherwave theremin and a vintage scientific X-Y plotter to create a hands-free drawing device that resembles an Etch-a-Sketch (Battle 2019).

The Bruel and Kjaer frequency analyser receives the audio from the theremin and converts it into two signals, one for pitch and one for amplitude. These two signals are sent via control voltage to the X-Y plotter which draws a line on paper, moving in response to the X-Y commands it receives via the control voltage.

LANGUAGE RESEARCH

As time progresses, society spends more and more time online. An existence inundated with text based language. Text that in its basis has been mostly unchanged for hundreds of years (Ferrara, Brunner & Whittemore 1991).

This text has served as an efficient way of expressing ideas and information, however when taken at face-value it can lead to misunderstandings and conflict (Mehndiratta, Sachdeva & Soni 2017). This may be due to the lack of many nonverbal components of communication that help to frame and contextualise spoken language (Bryant & Fox Tree 2005).

When communicating online it can be excessively complicated to deduce the tone of the speaker. Is a comment offensive or humorously sarcastic? Is someone joking in a self-depreciating manner or do they need help? Is the commenter saying something tongue-in-cheek or phrasing a question rhetorically?

In spoken language, these aspects can be conveyed by altering the way in which the sounds/syllables are spoken. Is it possible to translate this concept to digitally created text? If so, what would be the rules and representation of these inflections (Goulart et al. 2020)?

VERBAL COMMUNICATION

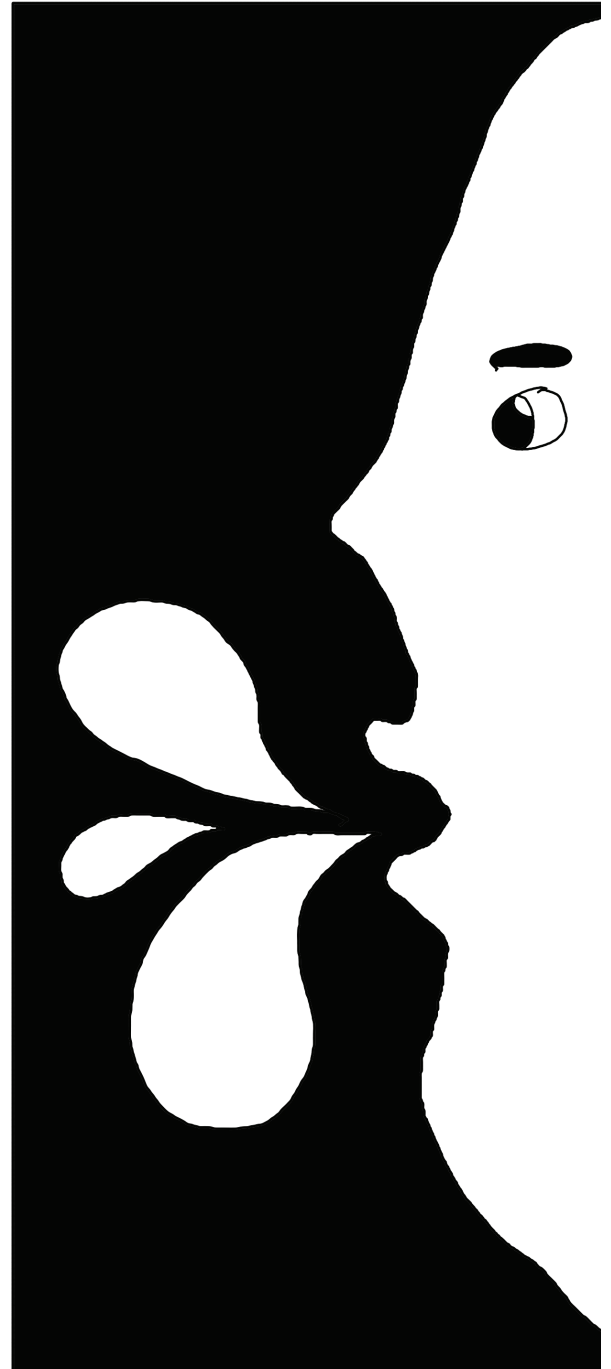
The structure of linguistics can be broken down into four main categories: grammar; discourse; style; and, lexicon (Key 2011).

Grammar is the formation and composition of sounds, words, phrases, and sentences that make up a given language (Hewes 1992). Discourse is the social practice of language, where the arrangement of ideas and phrases can set the genre and meaning of the speech (Toolan 2007).

The discourse and grammar of a language can adapt for specific purposes and practices such as changing from a casual to a formal dialogue (Shany-Ur et al. 2012). These formations of language - known as registers – are an important part of communication as they allow a person to assimilate to their environment and further encourage group identity (Goulart et al. 2020).

The lexicon of a language is the internal dictionary that a person has to form their communications (Rheindorf & Wodak 2019). It is composed of the known words as well as the additional structures such as prefixes and suffixes.

The style (or stylistics) is the complex intricacies of the formation of sentences and other dialogue aspects. It is what creates the emotional tones such as satire, rhetoric, or irony (Cheang & Pell 2008). It varies between person and person and is what gives a person's linguistics their personality.



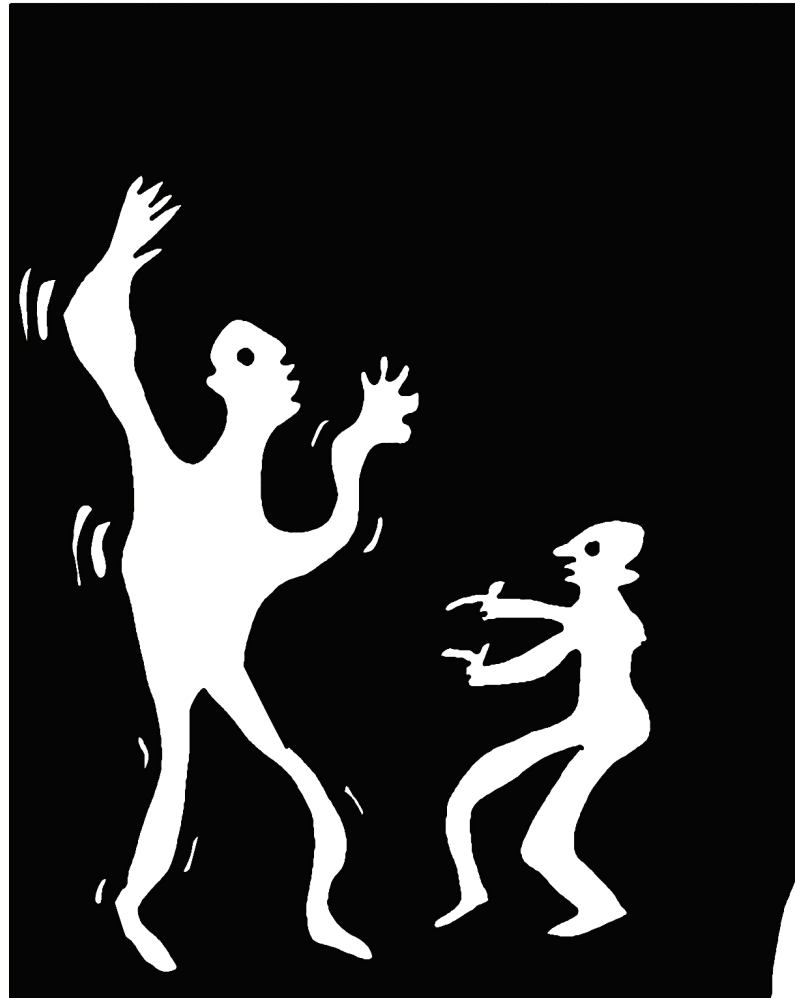
NON-VERBAL COMMUNICATION

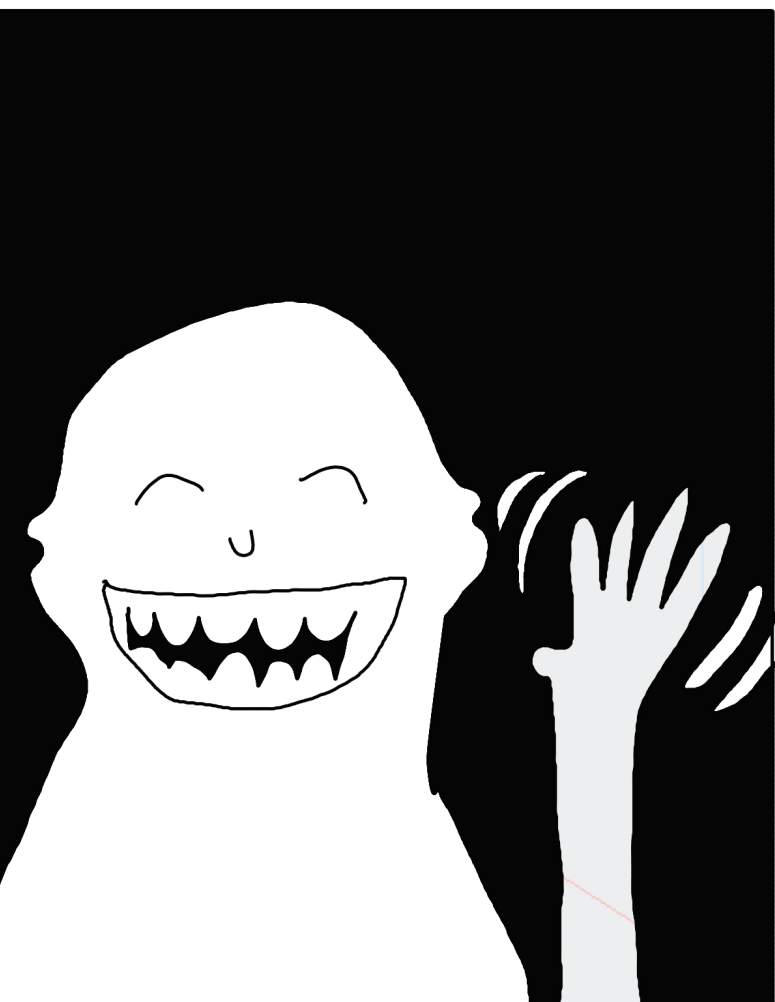
The ways in which a person moves and uses their body can convey and communicate emotions and information (Marmpena, Lim & Dahl 2018).

The meaning and intentions of speech can vary heavily based on aspects such as eye contact, posture, gestures, and other forms of body language (Kleinke 1986). In addition to these actions, communication can also be further modulated and expanded with alterations in distance (also known as proxemics), interactions with objects and the environment, and the use of touch (haptics)(Pennycook 1985).

The speech itself also has within it key aspects of non-verbal communication. Non-verbal voice aspects (paralanguage) as well as the use of time (chronemics) allow for the spacing of words, expression of thought processes, and provision of insight on the level of consideration of the speaker (Walther & Tidwell 1995).

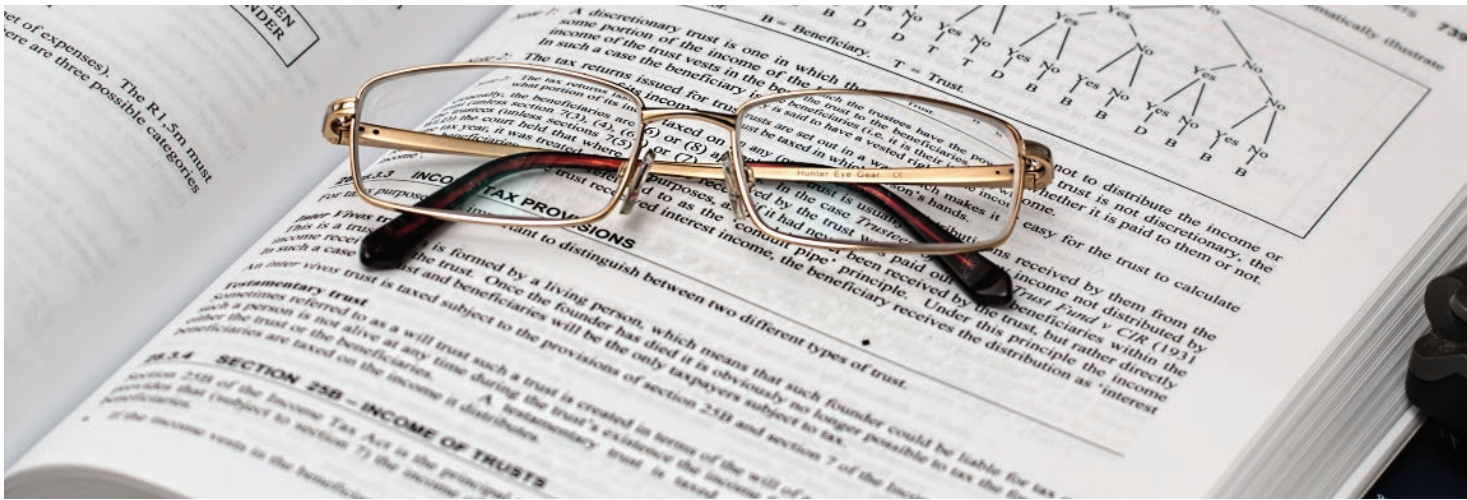
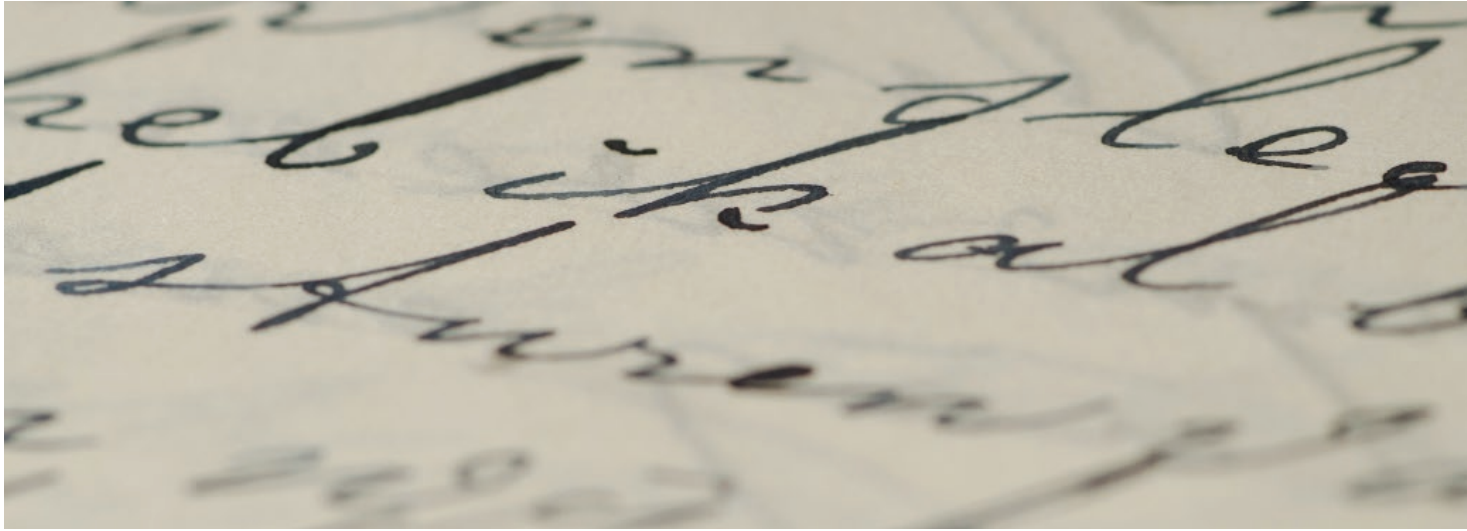
Paralanguage is an incredibly important part of speech (James 2017). By altering the pitch, volume, and intonation the user can add depth to the meaning of the words used in conversation. The use of paralanguage can be both conscious and subconscious and can even exist within handwriting and sign language in an altered form (Tsonos & Kouroupetroglou 2008).

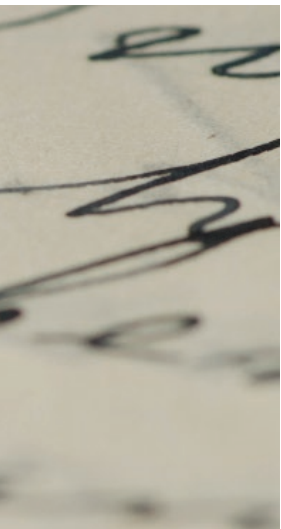




The prosodic features of language can shape the words and sentences, altering their meaning and importance to the listener (Hunston 2007). By altering the rhythm and stressing of syllables the communicator can alter the tone of the speech, incorporating emotion or contextual elements such as sarcasm or focus (Cheang & Pell 2008; Mehndiratta, Sachdeva & Soni 2017).

Gestural actions differ from other physical forms of nonverbal communication. They are a means of conveying a wide variety of information and feelings, allowing movements of the body to communicate their own messages or assist verbal speech - emphasising and modifying the meaning (Iverson & Goldin-Meadow 2005).





WRITTEN LANGUAGE

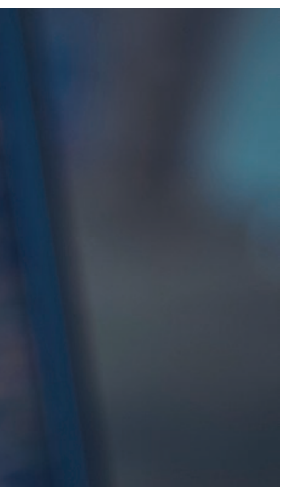
A written language (or literary language), through representative of the spoken language, can differ quite significantly (Toolan 2007). It is limited to the standardised symbols that represent sounds and information. In most cases written language has focused on specific registers and can even exist as a separate dialect to spoken language (Ferrara, Brunner & Whittemore 1991).



Language is standardised to allow for a shared meaning of the symbols and formation. This is to maximise the cohesiveness of a language within a community and to minimise misunderstandings in dialogue. The standardisation involves fixing the orthography, grammar, and dictionary. This can help to enforce cultural identity and allow for the transference of information accurately to a larger quantity of people over a greater period (Wang et al. 2019).

Written language primarily represents the phonemic structure of speech, with the composition of symbols representing the core sounds of syllables and words. There is also an inclusion of symbols that alter the grammar and punctuation of the language (Luangrath, Peck & Barger 2017).

Written language, especially in digital form is limited to the standardisation of its structure. As the form of the symbols (in their shape and distribution) have been allocated their near-fixed meaning and orthography, the medium in which they are presented highlight and focus only on these standardisations (Jensen 2014). This has led to the literary language being merely a representation of the strictly verbal component of spoken language. And even then, it in many cases exists with a separate array of registers and styles (Luangrath, Peck & Barger 2017).



These components of language, as stated before, are merely only components of a larger system of communication integral to human interaction.

TYPOGRAPHY

The way written language is presented can alter the way it is interacted with (Tsonos & Kouroupetroglou 2008). The formation of the text, in its shape and size and distribution can make the reader attribute different emotions to the text itself. It may also affect the readability of the text. In some ways it can be tied to the concept of a register in that it can shift the sociolinguistic tone and genre of the communication.

The shape of the text, such as the intricacies that make up a particular font, can infer certain characteristics, giving the text itself a unique voice (Keyes 1993). Certain fonts can also reference previously expressed identities such as those from history or pop culture (Spitzmüller 2019).

These fonts can then be presented with many variations, which include alterations to the proportions of the letters/symbols as well as the spacing, relative size and weight of the text (Castro et al. 2019).

The arrangement of the text within the medium itself may also influence the reader, whether through overcrowding a page or having it aligned in an obscure manner (Tsonos & Kouroupetroglou 2008).

TYPOGRAPHY EXPERIMENTS

Typography may have the means of representing non-verbal language through the alterations in shape and size. To assess the possibilities of the project, it was key to visualise how specific typographical changes affect the emotional context of the language as well as the representation of oral performance.

Which text looks like it was said slower?

Which text looks like it was said slower?

Which text looks like it was said slower?

Which text looks like it was said slower?

Which text looks like it was said slower?

Which text looks like it was said slower?

Which text looks like it was said slower?

Which text looks like it was said slower?

Which text looks louder?

Which text looks louder?

Which text looks louder?

Which text looks louder?

Which text looks louder?

Which text looks louder?

Which text looks louder?

Which text looks louder?

Which text looks louder?

How does the alteration in word height

WHICH TEXT LOOKS LOUDER?

change the way in which we view the

Which text looks louder?

movement of the speaker? Perhaps

WHICH TEXT LOOKS LOUDER?

changing the height of the text will alter the movement.

Which text looks louder?

Maybe changing how *much* the text leans

WHICH TEXT LOOKS LOUDER?

will make the *text seem like* it was spoken from a moving *speaker*.

If we combine an array of typographical changes on a per word basis, *would it* affect how we imagine the voice of the speaker? Will there be an added tone and personality to the text?

MESSAGE FORMAT

How should these concepts be presented?

The message itself is obviously of key importance. However, the medium in which it is presented also plays a large role. How differently it feels to read from a book than a phone. Or even on different forms of paper.

To remove the distraction of competing media and focus the user on the text as an artifact itself, it could be advantageous to have the created messages be displayed in printed form.

It was also decided that having a printed form that is narrow and long would highlight the temporal relation to long and short messages. It is also referential to classic digital message presentations such as Facebook messages and text messages.

To aid the text and to increase the emotional connection between messagers, an image of the sender should be present at the top of the message. This allows the recipient to see the sender's face and have it instantly associated with the message that is sent.

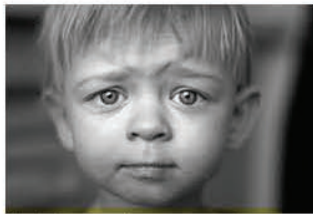
For the final artifact to become a full representation of the message, it will include the author's name and the time and date sent.

A printed form of this layout leaves a long strip of material that acts as a receipt of a message. To double down on this line of thought, the message should also be printed on thermal receipt paper.

This use of a thermal printer will have the benefit of novelty and an iconic experience as the printing has an audible sound. The printouts require no ink and the rolls of paper will last for many messages.

The messages will also remain linked until torn, forming a timeline of conversation which can in turn enhance the emotional significance of the artifact.

This is roboto 12 pt
 This is roboto 10 pt
 This is roboto 9p
 This is roboto 8pt
 This is roboto 7pt



This is robo-
 to 20pt
 This is roboto
 18pt
 This is roboto
 16pt
 This is roboto
 14pt
 This is roboto 12 pt
 This is roboto 10 pt
 This is roboto 9p
 This is roboto 8 pt
 This is roboto 7pt



8 bit

ABCDEFGHIJKLMNOPQRSTUVWXYZ
 0123456789@
 !?\"#\$%&'()*+,-./:;<=>
 ~`{|}~^_~:~X~Y~Z~\$~@~
 ABCDEFGHIJKLMNOPQRSTU
 VWXYZabcdefghijklmnopqrstuvwxyz
 0123456789@
 !?\"#\$%&'()*+,-./:;<=>
 ~`{|}~^_~:~X~Y~Z~\$~@~

1982 Computer Research Ltd

ABCDEFGHIJKLMNOPQRSTUVWXYZ
 LMNOPQRSTUVWXYZ
 WXYZ

abcdefghijklmnopqrstuvwxyz
 mnopqrstuvwxyz
 xyz

0123456789

\$ 00000000* 000% 0"'



Date: 21 May 2021

Time: 10:27:48 PM

Message sent by Ali

Hey Michelle, I

am **working** on the
audio recognition
 process right now.

My **LOVELY**
 housemate has just
 made a HUGE

burger

for me and some chips.
 I

am gonna

eat that. Then JUST
 AFTER I'll

go to the gym with
FULL STOMACH. I
 will let you know
 when I

am ready back
 home to have a video
 call **about**
the project.

See you then

Secatibe runtum earum sincilla velenimus autu nobit raecus di omniet

PRINTOUT PROTOTYPE

Using the format stated previously and following the limitations of a thermal printer (1 bit printing, 384px wide), prototype message printouts were made to visualise the potential outcome of the CHAT device.



Message sent from Malte
Sent on;
Date: 08/06/2021
Time: 12:45:13 PM

WOWEE. These
guys' work and
presentation design are
top notch.
Don't you **think Tron**?



Message sent from Chuan
Sent on;
Date: 08/06/2021
Time: 12:46:56 PM

Yes **Yes** Yes **Yes** Yes it
is Malta.
These
cool cats are my
babies. They are going to go
far
for sure. **H. D.** All round.
I can't **wait** for them to finish
though
so that I can make myself a
cheese sandwich. **Yum** yum in
my tum tum.



Message sent from Jen
Sent on;
Date: 08/06/2021
Time: 01:10:05 PM

These guys are **absolutely**
the bee's knees. I can't **believe** you
got to be in the studio with
these legends for a whole
semester. I can't **wait** to see this design
project get developed further. I would **love**
to have one of those artefact's printing out
what I say exactly the way I would say it.



Message sent from Chuan
Sent on;
Date: 08/06/2021
Time: 02:01:28 PM

I hear you **Jen**.
I hear you. These guys have far
exceeded my expectations.
Okay okay in a **min-**
ute.
Sorry, my dog wants to go
for a walk. See ya.



Message sent from Emma
Sent on;
Date: 08/06/2021
Time: 02:36:45 PM

O h. **Wow**. These guys are
great. I can't wait to see team
chats design work at the NGV soon. And
hopefully and eventually in my living
room. Give **these guys** the **highest**
mark possible **Tron**.



Message sent from Ross
Sent on;
Date: 08/06/2021
Time: 02:46:25 PM

That's
pretty good guys
Could you guys send **me**
this video through email?

TECHNOLOGY

To create a text based analogue of speech, it would be ideal to first start with the audio recording of the speech itself. This could then be translated into the typographical form.

The speech would also have to become the written text so that there is only a need for one input of information. This will require a microphone input and computing hardware that can perform speech to text conversion.

As stated in the previous section, it would be ideal to have an image taken, therefore a camera that can easily be intergrated into the computing hardware is required.

There is also a need to measure the participant's body movements around the device because many of the non-verbal characteristics of interpersonal communication are related to a person's overall position in space and movement within their environment.

Finally, the device will need to have a thermal printer to print off the message "receipts".

[Not included in this prototype is the means of sending the message. It could either be through radio via an adafruit feather or through traditional internet communications via wifi.]

COMPUTING

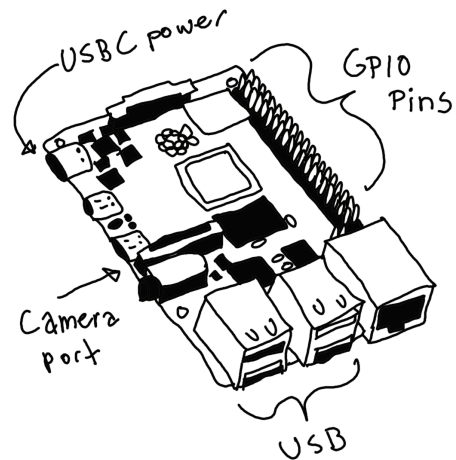
Because of the potential need for complex processing of data and integration of several technologies within one processing pathway, it was considered as optimal to choose a capable single board computer such as the Raspberry Pi 4.

This board has accessible GPIO pins that can connect to devices such as the thermal printer. These GPIO pins can power and communicate with the printer over serial communication, sending printing commands and receiving information on the printing state and paper levels.

The Raspberry Pi 4 has complementary camera modules that can attach via a specific camera port. This is paired with simple software integration and control.

The device itself runs a fully capable linux based operating system that is easy to setup and program within its environment. The device can be accessed remotely to allow for ease of programming/development.

The majority of the programming will be using the language of Python as it is incredibly versatile and libraries exist for the control of the needed components. Detailing of this will be shown later in the document.

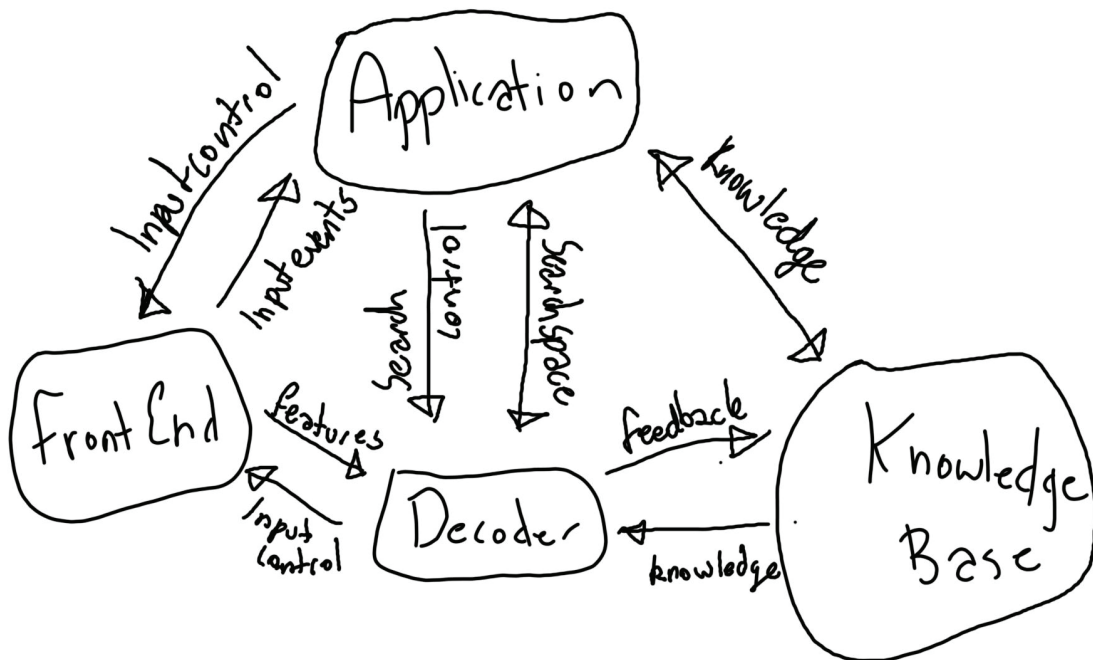


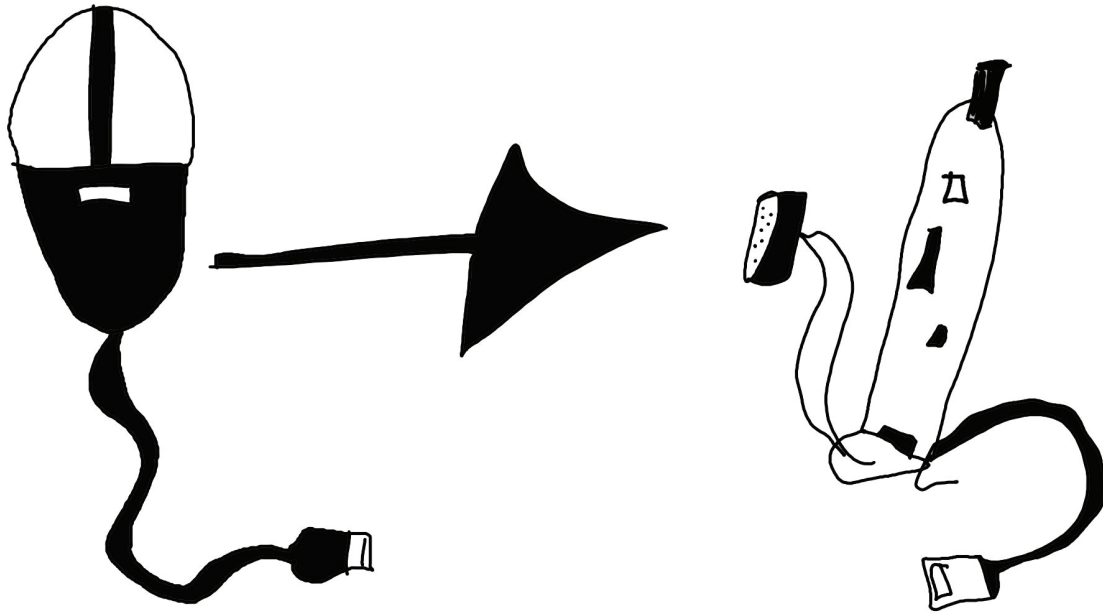
SPEECH TO TEXT

For the speech to be converted into text, a speech to text engine is needed. There are several available to use including Google Speech-To-Text, Microsoft-Cognitive-Services and IBM Watson. The one chosen however, is CMU Sphinx's Pocketsphinx. This is an open source toolkit for speech to text recognition and has easily accessible code for implementation in private projects such as CHAT.

This speech to text toolkit can be used both for live speech recognition as well as reading a previously recorded audio file. For this project, an audio file is preferable as it would enable a seamless recording of the audio and other variables such as body movement (Prazdnichov 2018).

Pocketsphinx also has the ability to show the start and end times of the words. This means that it is possible to determine how long each word took to say as well as when they were said.

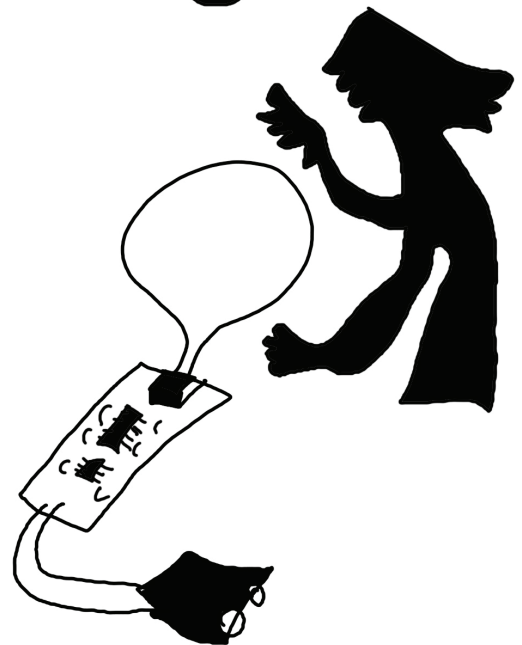
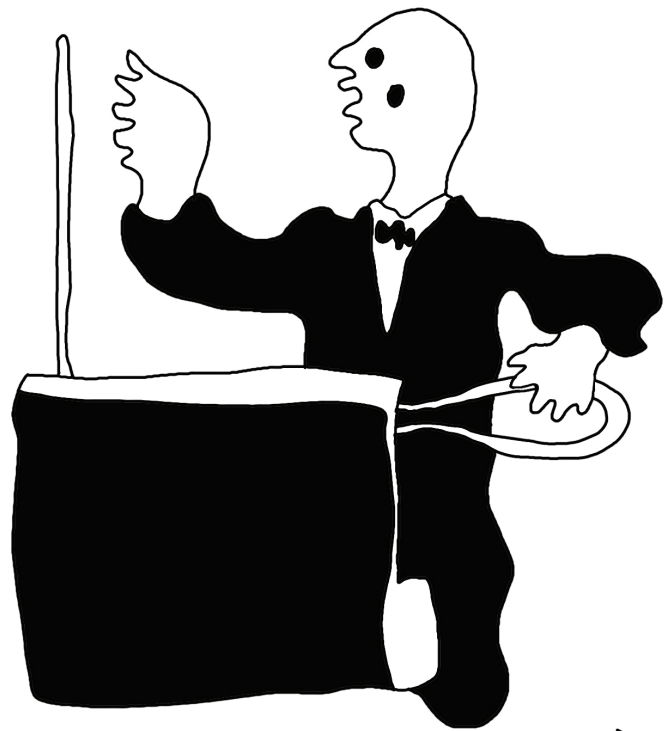
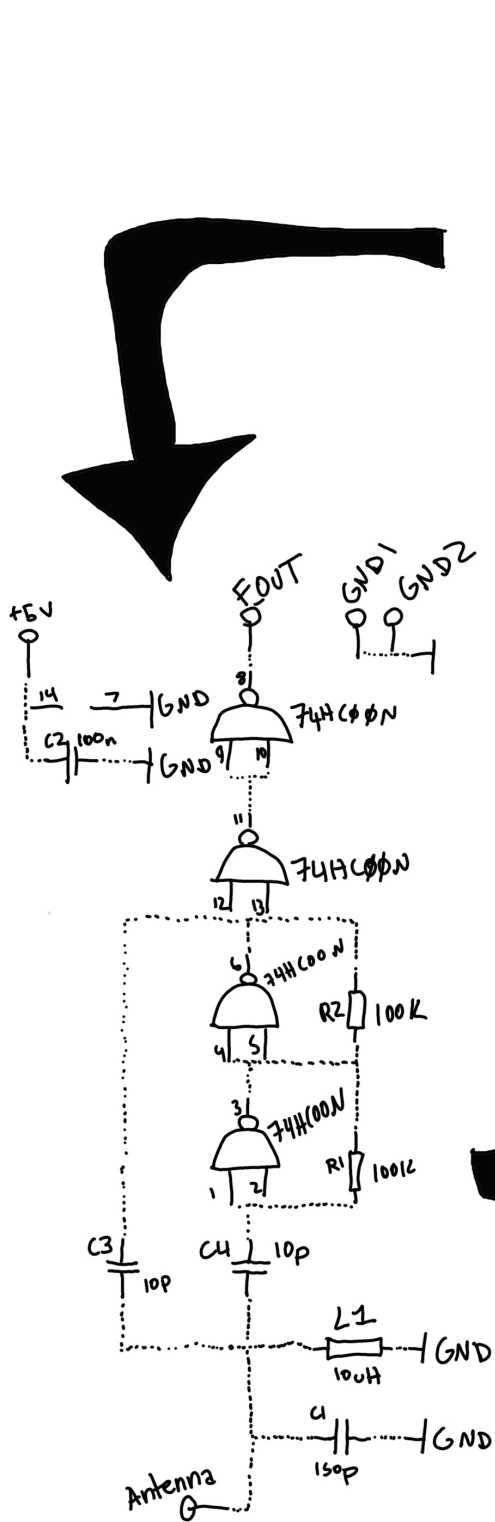




M I C R O P H O N E

For the audio input, a usb microphone (Audio Technica's AT9934) was gutted and the internal componets were repurposed for the CHAT device.

The USB microphone requires minimal configuration to implement in the Raspberry Pi and allows for minimal latency. It has an internal sound-card which provides the gain control and analog to digital conversion so that the Raspberry Pi merely needs to receive the digital signals and intergrate them into the speech to text code.



ELECTRO - MAGNETIC FIELD SENSOR

Omnidirectionally sensing a body is an uncommon need in modern technological devices. There is not a wide variety of technologies and methods available to sense from a distance how a person moves in three-dimensional space.

There is however, a musical instrument which is controlled in a manner that mimics many of the gestural movements of a speaker in conversation. This instrument is the theremin.

Invented in the 1920's by celloist and physicist León Theremin whilst he was developing proximity sensors, the instrument is able to detect and react to the human body when it was near two radio antennas, resulting in an alien-like violin sound.

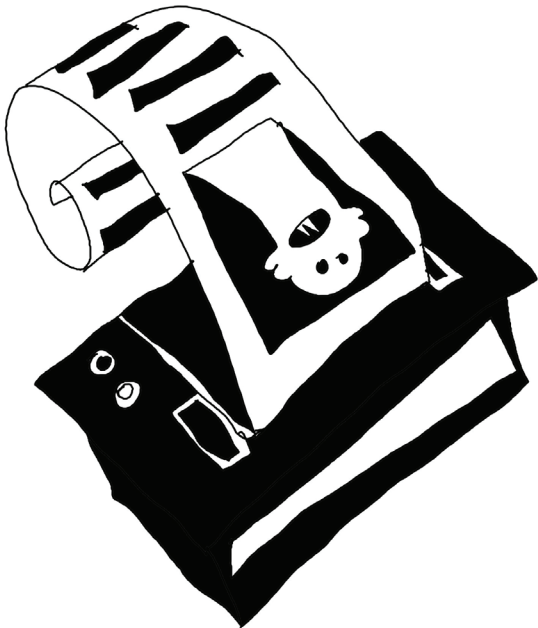
This instrument has been developed since, with many iterations, fine tuning and simplification.

Though the theremin uses two antenna to control pitch and amplitude of the sound, the CHAT device will only need to monitor one overall body instead of the user's hands. This reduces the needed antenna by half and greatly simplifies the circuit.

From the now simplified circuit, a prototype was able to be built that can send the measurements of a person's movements to the Raspberry Pi.

A circular antenna design proved to yield optimum results and could respond to movements up to 2m away.

THERMAL PRINTER



The printing of the message "receipt" was chosen to be from a thermal receipt printer. This is for many reasons:

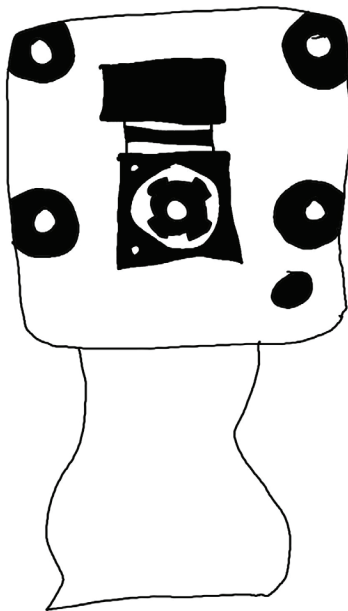
Firstly, a receipt printer has the ability to print out long rolls of monochromatic content. This creates a shape of text and imagery that is visually similar to that of a modern digital text message such as that from a SMS or Facebook message.

Secondly, the printer itself uses no ink and instead has paper that reacts to a heating element inside the printer. This minimises involvement in upkeep and maintenance of the device as well as reducing the cost of continued function.

The printer also has a iconic performative aspect as it audibly prints line by line, jutting the paper out in small bursts. This combination of sound and movement can aid in the emotional connection with the device and the artifact that it creates.

The specific printer chosen was the Adafruit Mini Thermal Printer. This printer has preexisting libraries and code for use with a Raspberry Pi.

CAMERA



Due to the selection of a Raspberry Pi as the computer of choice, the optimum camera module would be the Pi Camera. This camera has a large level of support for software and hardware development. Libraries are easily available for the Python language and access to the camera's controls are embedded into the default Raspberry Pi software. The Raspberry Pi also has a specific port that allows for direct connection to the camera module.

BODYSTORMING

To understand how the CHAT device is to be interacted with and what affordances are needed in its design, the method of bodystorming was implemented. This is the act of using roleplay to walkthrough the user interaction of the device.

The CHAT team began with a small cube that was the approximate size needed to contain the components of the device.

We then ran with the narrative of coming home and seeing the device. What would be our intent when we first see it? How would we know if we have a message? How visible should it be?

After walking over to the device what do we do? From the roleplaying, it was apparent that some of us had the urge to pick up the device to inspect it. After inspecting CHAT, it was observed that there was a significant need to see what was the front and back of the device before we were able to continue the interaction.

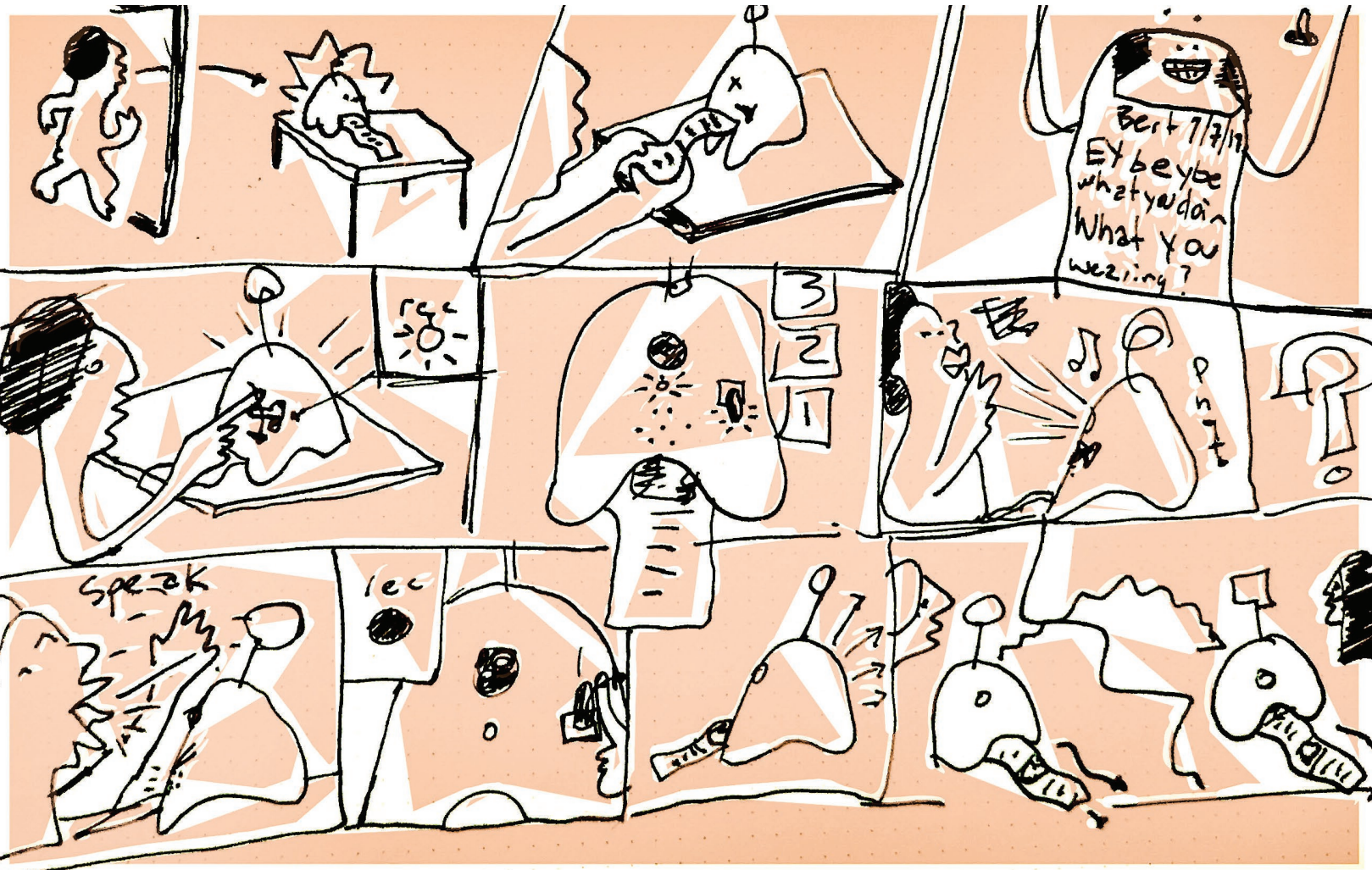
The box was then modified to have a visible camera at the front.

After many runthroughs it was deduced that the optimal structure of the device would have all initial interfaces visible on the front, the EMF sensor on the top and the printer on the underside. This was to establish a visible hierarchy and order of interaction with the interfaces.





STORYBOARD



The CHAT device is planned to be used as follows. The user enters the room and sees the device. They then either see a printed message or just the device itself. If there is a printed message they will then read and flick the switch to begin the message recording process. If there is no message then they will just flick the switch to begin.

After switching the device on, the user will pose for their photo and the recording will begin. The user will speak until they have finished their message, moving freely and naturally throughout. They will then flick the switch off and the device will print their message and send it to the recipient.

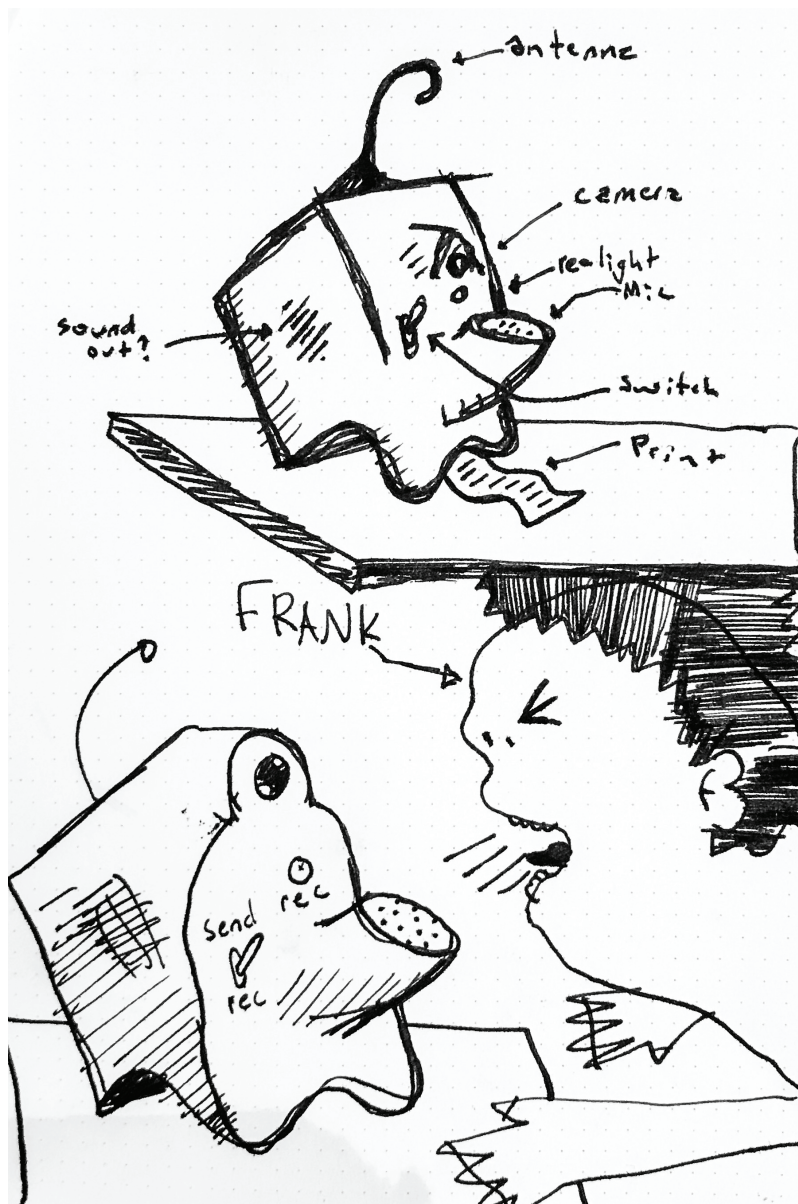


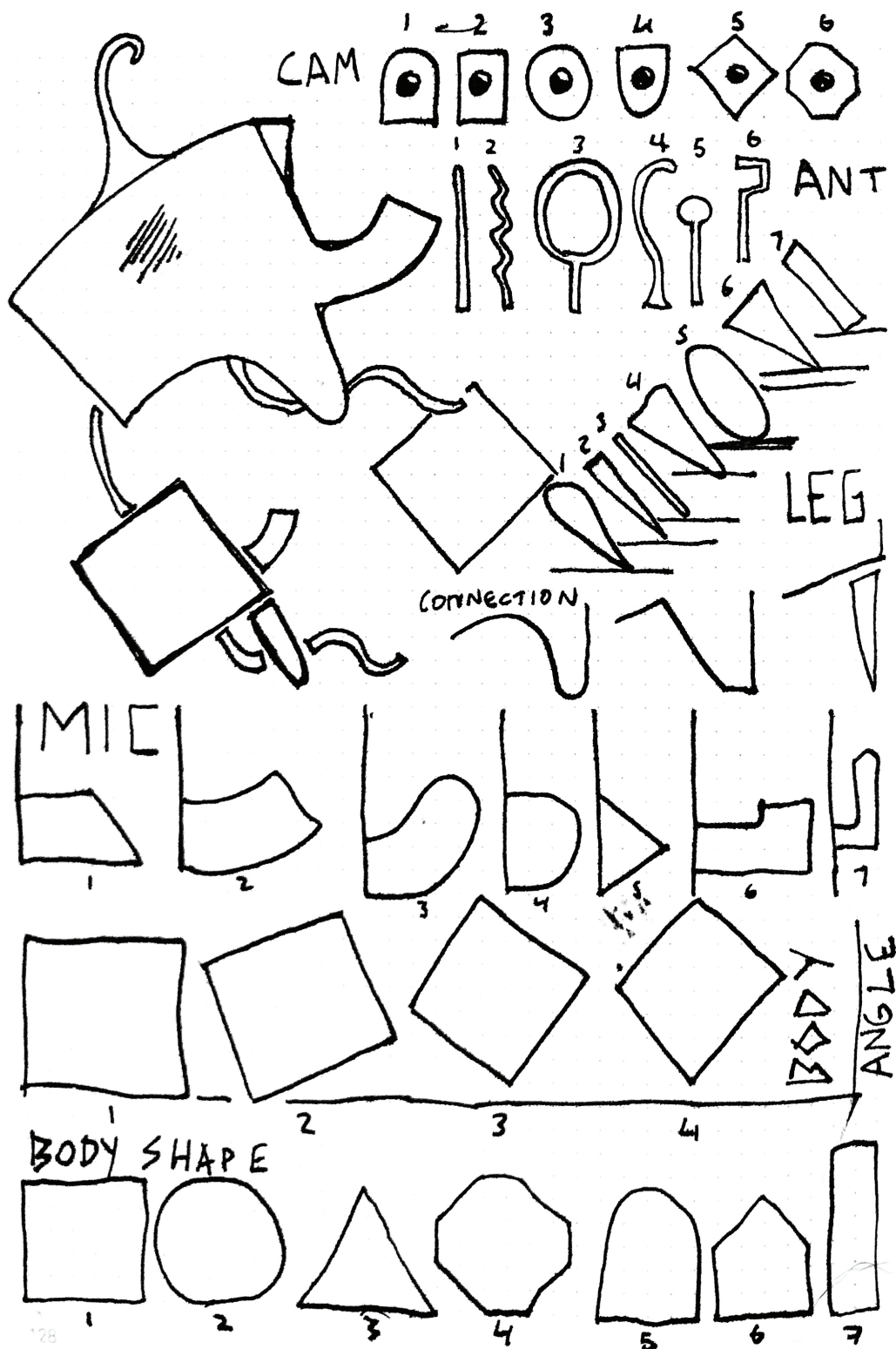
PHYSICAL STRUCTURE

After the bodystorming exercise it was essential to begin the exploration of shape and structure.

The device should be easily picked up and confidently interacted with. These qualities require the device to be small, light, visually sturdy and durable. To achieve this, it was decided that the body would be initially 3D printed. This also allows for the device to have a more abstract shape if necessary.

The design began with a cuboidal shape which was rotated, lifting the front of the device up. This angles the printer to ensure that the receipt will project towards the user. The camera, microphone and switch will be front facing. The microphone will have a shape that signifies its purpose and the camera will be angled to counter the body rotation.





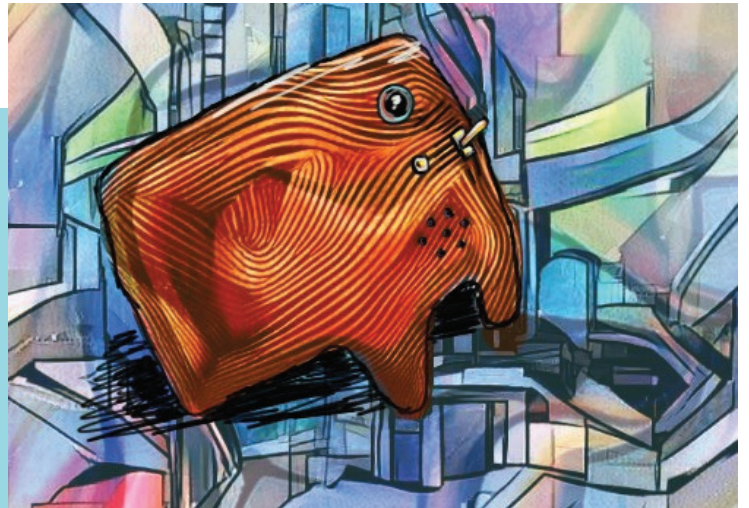
The specifics of each physical component was refined to simplify the overall aesthetic and form the device into one cohesive unit. Shaping the body with soft curves make it inviting to touch whilst angling and protruding components emphasise purpose and function. The angling of the device prevents the antenna from protruding towards the user's face and also angles the microphone towards the mouth without the need for additional protruding shapes. Though above, you can see potential microphone structures, it was decided that the presence of a patterned array of holes was only affordance needed.



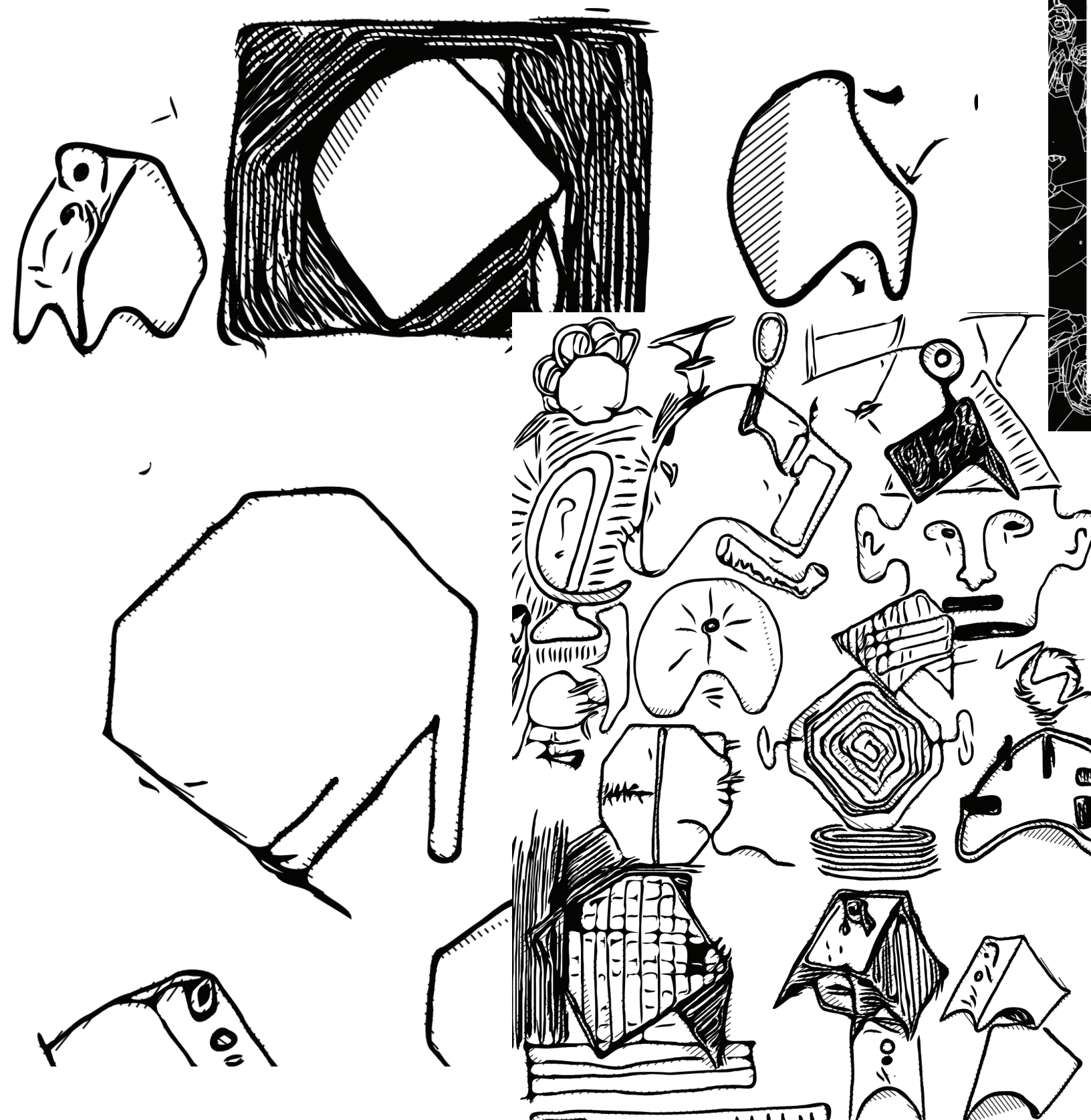
DESIGN DECISIONS

The design of the physical structure was further refined with experimentations in childish zoomorphic structures and the stripping back of all unnecessary shape.

The form could be angular and sharp or soft and playful. Below is an illustration of what was seen to be the minimal level of shape. This has the angled body and aimed printer with only a few curves. The only protrusions are the legs and the toggle switch. It now presents itself like a small alien creature. Something that could even be considered endearing or cute.



FINAL SKETCHES

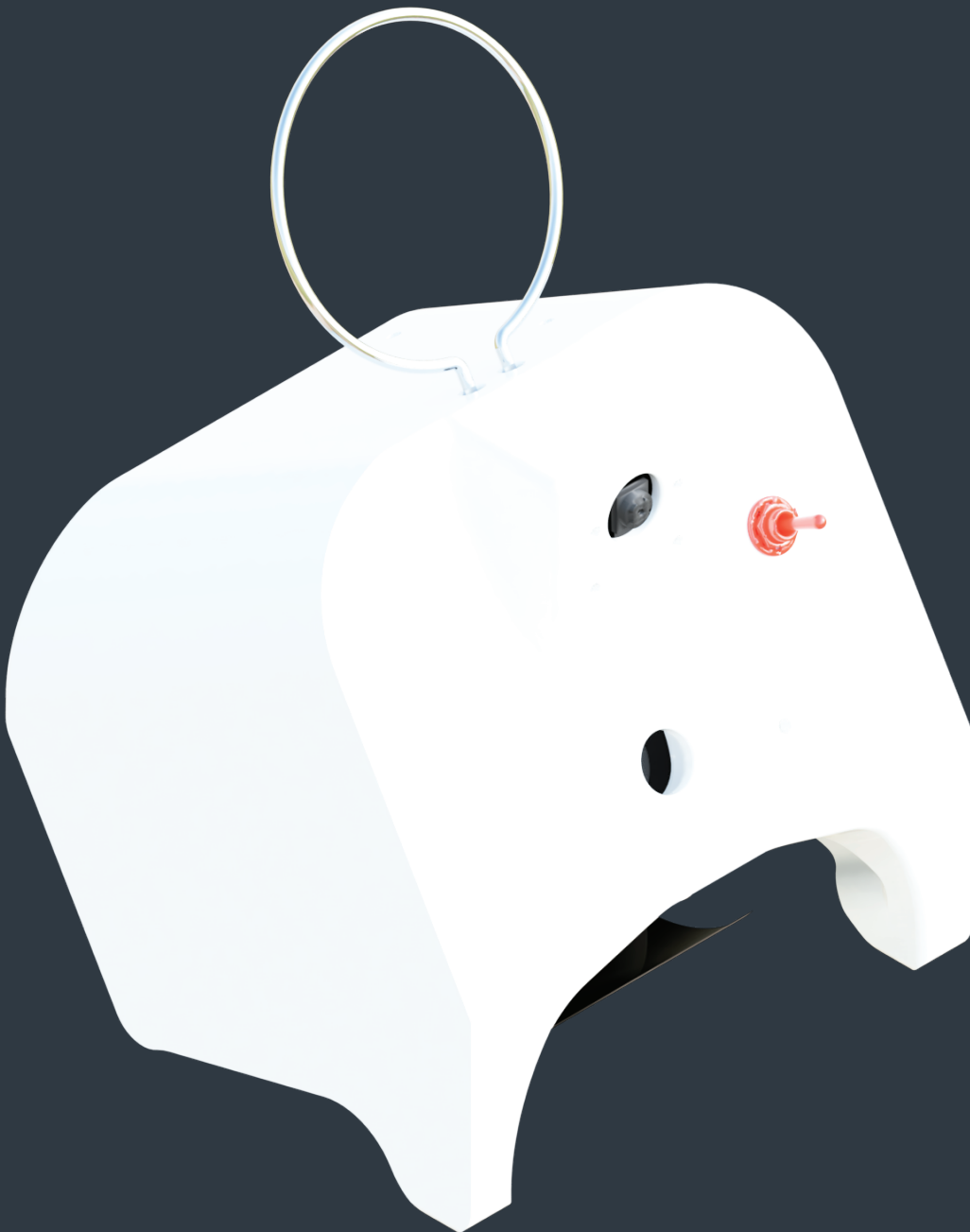


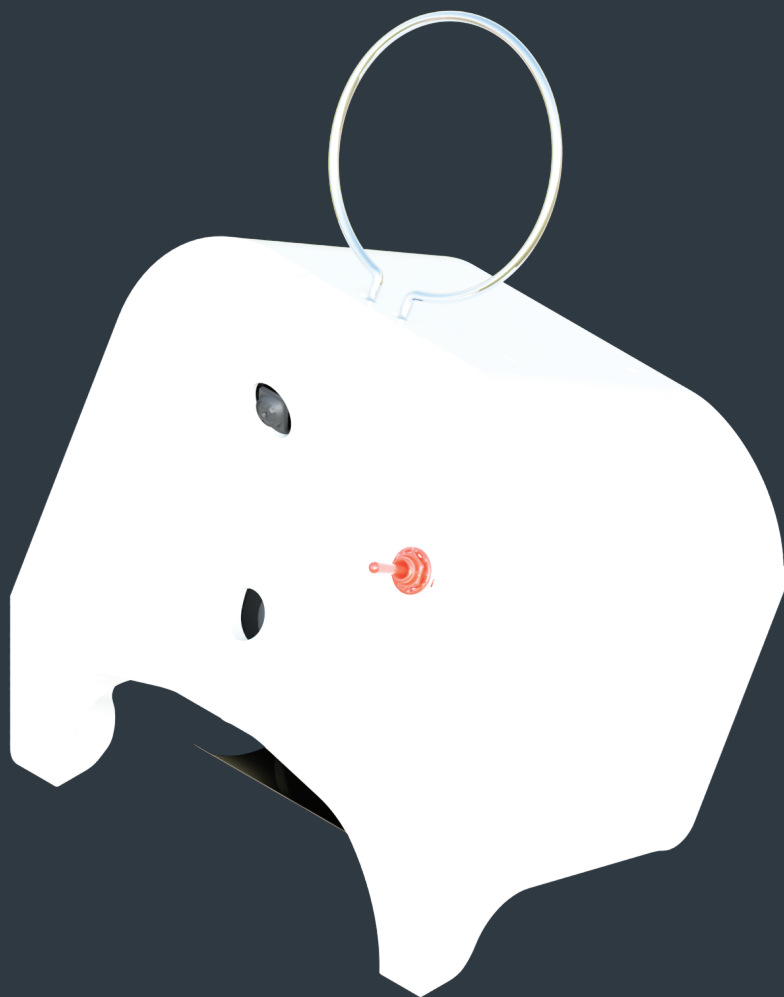


Further refinement and progression in the technical aspects of this project lead the design to become that presented above. Keeping the simple affordances and signifiers and overall soft zoomorphic shape, the device is a balance of smooth curves and simple geometry. A square front face allows the user to easily identify the front and the back of the device. The camera is angled to run parallel with the surface that the device is sitting on. The front legs remain soft so reaching between them will have no level of intimidation or caution. The electromagnetic field (EMF) antenna protrudes proudly from the top, easily visible but not at all in the way.

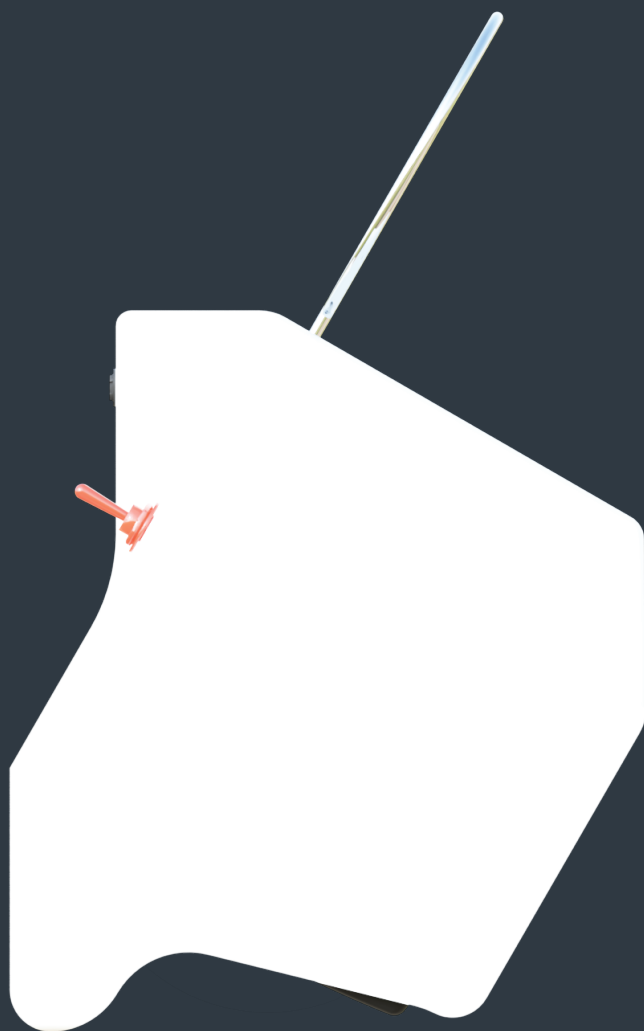
A small size and curved back allows the device to be easily picked up and played with.

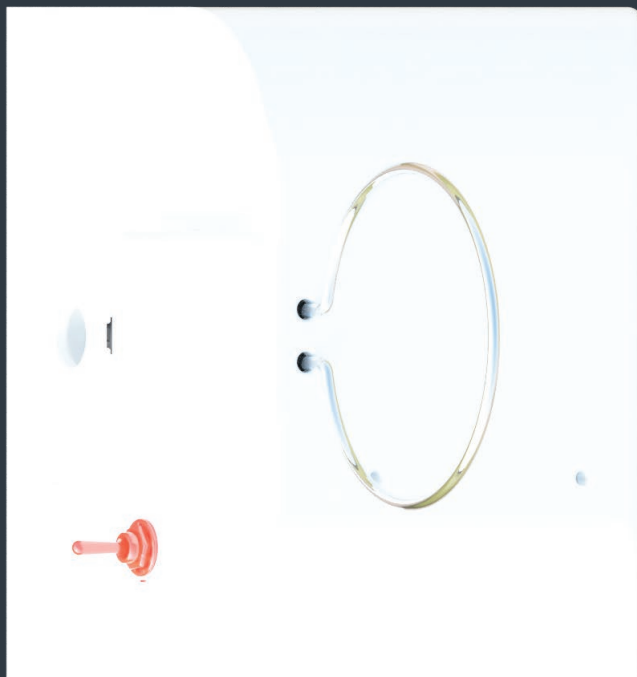
FINAL DESIGN

















3D PRINT





The body was printed in two components. This allows for a simple manufacturing process. The flat sides and gentle curves minimised the supporting material needed to print, thus reducing the cost.

The printer slides into the base of the structure and the Pi is mounted on the lip above it. The front has screw holes for the camera and the microphone PCB is slotted into the formed crevice and glued.

The two pieces slot together, held tightly with screws through the aligned holes.



The printed body is light-weight and convenient. easily played with and able sit on any surface. It can even balance on your head!



PROGRAMMING CHAT

OVERVIEW

For the CHAT device to convert the speech to a modulated typographical printout, the programming language Python was chosen. This was because the language has a large level of support for the Raspberry Pi as well as speech to text libraries.

It was also determined, that for simplicity of function, as well as the optimisation of processing resources, the programming will be contained within one continuously running script.

This program needed to first react to the change of switch state, so that it knows whether CHAT is set to send or receive signals. This changing of state to ON will initiate the taking of a photo which will be saved for later use. After this image is taken, the device will begin recording both the audio input and the input signals from the electromagnetic field (EMF) sensor. These will continue recording until the user flicks the toggle switch into the OFF position.

Once the switch is flicked, the recording of the EMF and audio finishes and the readings are saved to be used in the next steps.

The speech to text engine (CMU Sphinx) begins and analyses the audio file,

extracting the words from the speech with their start and end time.

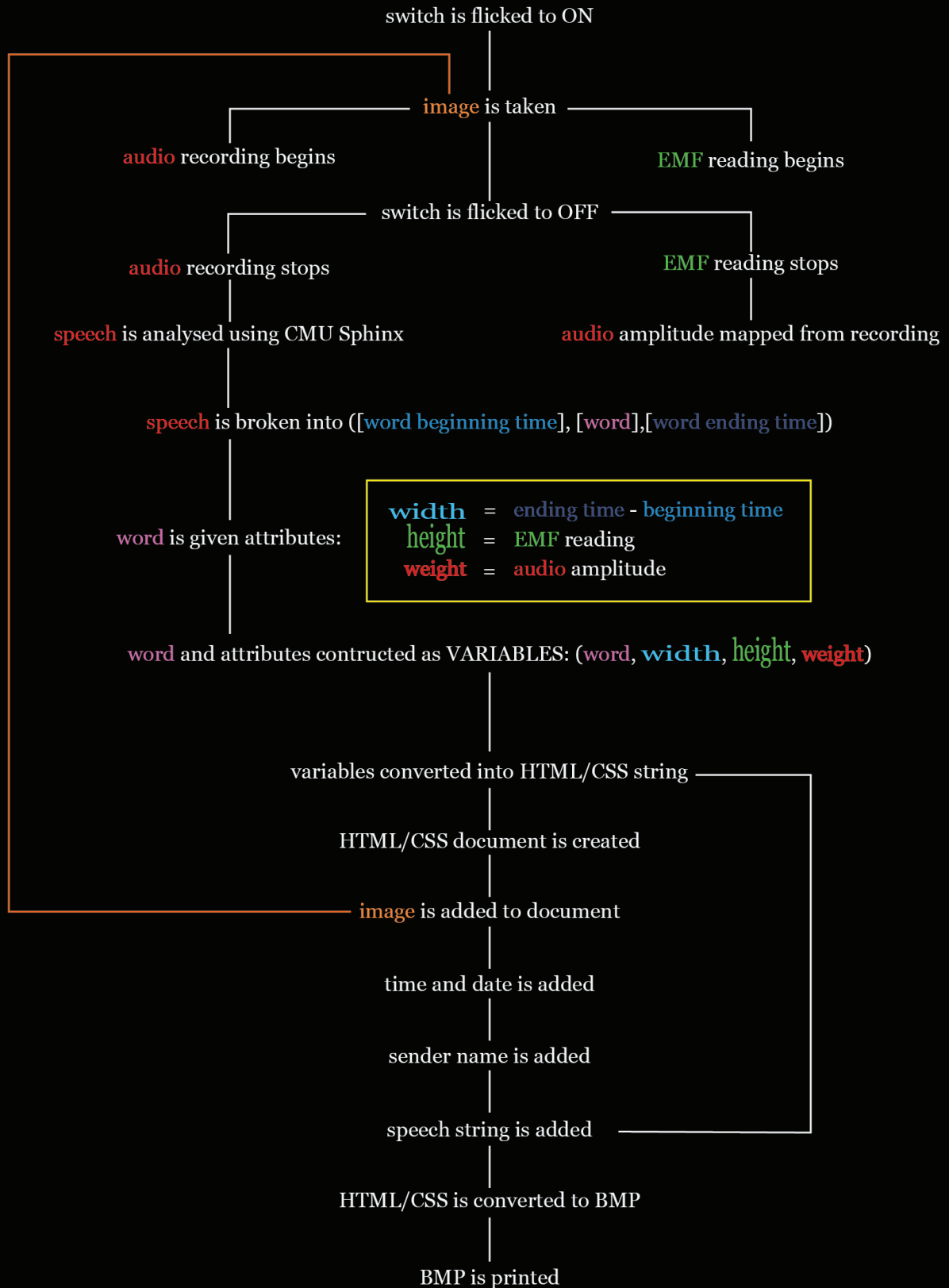
These start and end times are then used to break the EMF reading and audio file into segments which then have their average amplitudes extracted.

The word is then given attributes from the extracted values with the time taken for a word to be said becoming the word width variable. The EMF reading will become the height variable and the audio amplitude of the audio recording will become the weight of the text.

The next step is to use the words and their variables to create an HTML string using CSS styling to create the typography.

Firstly however, the HTML file will be created, the image previously taken will be added to the beginning of the file, then the time and date as well as the sender name will be added underneath. The speech will then be created with the variables modulating typographical parameters.

The HTML file is then rendered as a BMP image file which can then be sent to the thermal printer.




```

#Speech to text with more

import RPi.GPIO as GPIO
from pocketsphinx import AudioFile
from datetime import date
from time import sleep
from picamera import PiCamera
import os
import imgkit #https://github.com/jarrekk/imgkit

#printer setup
import serial
uart = serial.Serial("/dev/serial0", baudrate=19200, timeout=3000)
import adafruit_thermal_printer
ThermalPrinter = adafruit_thermal_printer.get_printer_class(2.69)
printer = ThermalPrinter(uart)

#audio file setup
import pyaudio
import wave
data_path = "/Desktop/audio"
file = "rec.wav"
with open(os.path.join(data_path, file), "r") as f:
    content = f.read()

#TimeGet
td=datetime.now()
td=td.strftime("%d-%m-%Y_%H-%M-%S")

```

Code Setup

The code begins with the importing of needed libraries and functions (see above). To simply access the GPIO pins on the Raspberry Pi so that the toggle switch can be used, the RPi.GPIO library is called. The subsection of the pocketsphinx speech to text library is imported. Time is imported so that the time and date can be added to the printout. The picamera library is imported and then os which allows for easy file management. The printer is then setup by importing serial which allows for serial-
port communication to the printer.

The communication is then configured and the specific library of communications to the printer is imported.

The audio file is then configured by firstly importing the pyaudio and wave libraries which then allow for an audio file to be created in a specific folder thanks to the help of os. This file is then opened and able to be edited within the Python code.

Now a variable is set to be the time and date.


```
def cam():  
    #Camera  
    camera = PiCamera()  
    camera.resolution = (1024, 768)  
    # Camera warm-up time #to be implemented as 3 sec countdown  
    sleep(3)  
    #capture and resize to good size. save as bmp just in case.  
    camera.capture('selfie.bmp',resize=(320, 240))
```

Camera Function

A function is created for the camera module. This is so that whenever needed, this section of the code can be called and an image will be taken at the needed resolution. It uses the previously imported picamera library and accesses the camera through the camera module port on the pi. The camera's resolution is 1024 by 768 pixels however the printout will be only 384 pixels wide so when the image is taken it will be resized to 320 by 240 pixels. The image is simply named selfie.bmp and will be called later in the code.

```

def rec():
    now = datetime.now()
    dt_string = now.strftime("%d-%m-%Y_%H-%M-%S")
    form_1 = pyaudio.paInt16 # 16-bit resolution
    chans = 1 # 1 channel
    samp_rate = 44100 # 44.1kHz sampling rate
    chunk = 4096 # 2^12 samples for buffer
    #record_secs = 60 # seconds to record (if limit is needed)
    dev_index = 2 # device index found by p.get_device_info_by_index(ii)
    wav_output_filename = file # name of .wav file

    audio = pyaudio.PyAudio() # create pyaudio instantiation

    # create pyaudio stream
    stream = audio.open(format = form_1, rate = samp_rate, channels = chans, \
                        input_device_index = dev_index, input = True, \
                        frames_per_buffer=chunk)

    print("recording")
    frames = []

    # loop through stream and append audio chunks to frame array
    while GPIO.input(10)==True:
        data = stream.read(chunk)
        frames.append(data)

    print("finished recording")

```

Recording function (BEGINNING)

The recording function (called later by reading the switch as being ON) begins by calibrating the audio input to the sample rate and channel number that is ideal for the speech to text processing. The input device is set to device 2 which is the usb microphone, and the recording filename is set to that which was created earlier.

The pyaudio system is started so that these calibrations can be implemented and the recording begun. A recording stream can now be started with the calibrations and an array of recorded audio values (frames)

will be created to be filled.

The recording will then continue while the toggle switch (GPIO.input(10)) is still in the ON (==True) position.

```

# stop the stream, close it, and terminate the pyaudio instantiation
stream.stop_stream()
stream.close()
audio.terminate()

# save the audio frames as .wav file
wavefile = wave.open(wav_output_filename, 'wb')
wavefile.setnchannels(chans)
wavefile.setsampwidth(audio.get_sample_size(form_1))
wavefile.setframerate(samp_rate)
wavefile.writeframes(b''.join(frames))
wavefile.close()

#html settings
filename = "Your Message.html"
pagewidth = 384
pageheight = 10000
dpi = 72*(1)

fps = 100

#toggle on/off switch
GPIO.setwarnings(False) # Ignore warning for now
GPIO.setmode(GPIO.BOARD) # Use physical pin numbering
GPIO.setup(10, GPIO.IN, pull_up_down=GPIO.PUD_DOWN) #on/off switch on pin 10

```

Recording function (END)

When the switch is no longer in the ON position (`GPIO.input(10)==False`), the stream will stop and then close and the audio system will terminate. The audio file will then be opened and calibrated to the audio settings and then filled with the recorded audio frames. This file is then closed and can now be used for further processing.

HTML and Toggle Switch Calibration

The HTML file is set to the name "Your Message.html" which will exist in the root folder. It will have a page width of 384 pixels as it is the width that can be printed. It has a maximum length of 10000 pixels at this stage. The DPI is set to 72 so that font sizing within html is directly converted to the pixels of the print. The FPS that is set to 100 is the effectiveness of the speech to text. This setting is in the low to medium range. The GPIO is calibrated for the toggle switch so it can be read as stated previously.

```

if GPIO.input(10)==True:
    cam()
    rec()

    f=open(filename, 'w')
    image = 'selfie.bmp'
    #make the whole layout
    result = "<!DOCTYPE html><html><head>\n
    <style>\n
    body{{padding:8px ;margin: 0 auto;width: 384px;background: #FFF;font-family: \
    \"Georgia\", \"Calibri\", monospace;}} \
    h1 {{ padding: 8px;margin:0 auto; width: 384px; background: #FEE;\n
    text-align: center; font-size: 260%; font-family: \"Georgia\", \"Calibri\", monospace; }}\n
    </style></head>\n
    <img src ={} alt ='selfie'>\\n \\n <h1>{}</h1> \\n\n
    <h1>Name of Sender</h1>\n
    <body>".format(image, td)

    print("processing has begun")

    for phrase in AudioFile():
        sent=""
        for s in phrase.seg():
            word = s.word
            start = s.start_frame
            end = s.end_frame
            tlength = end-start
            clength = len(word)
            height = tlength/clength
            width = clength
            weight = 1
            if word == '<sil>':
                word = ' '
            elif word == '</p>':
                word = '.'
            elif word == '<s>':
                word = "<p>"

            #w = [word, tlength, size]
            if len(word)>1 and word[0]!=' ':
                w = '<span style="font-weight: {}; font-size: {}; font-stretch: {};"> \
                {} </span>'.format(weight, height, width, word)
                sent+=str(w)
                sent+=" "
            result+=str(sent)
        else:
            result+="  
</body></html>"

    #options for html to image
    options={
        "--format": "bmp",
        "--page-width": "384"
    }

    imgkit.from_string(result, 'message.bmp', options)

    #save html for whatever reason
    f.write(result)
    f.close()
    #wait 2 sec
    time.sleep(2)

    printer.printBitmap('message.bmp')

```

Speech to HTML to BMP Image

This is the core part of the code. It is where the previously recorded audio will be processed using text to speech and then converted into an HTML webfile with CSS styling. This can then be converted into a BMP image that is able to be printed by the thermal printer over serial port communication.

This section begins if the toggle switch is read as being ON. It then begins the camera function (previously described) and then the audio recording function.

Once these functions have finished (after the toggle has been switched to OFF) the HTML file will be opened and the taken image will be set to a variable and stored in memory. A string is created (result) that is filled with the beginning of an HTML document that includes the CSS formatting of the page and the image taken. It will also include the name of the sender(to be added) and time/date.

The string at this point ends by opening a <body> section of the HTML file. This is so the next section (the speech to text) can easily input the words to be presented in the required format.

This format is as follows: '{} '.

 allows for CSS styling to be implemented on a per-word basis, the font-weight will adjust the thickness of the text, the size will adjust overall height of the word and the stretch will adjust the width. Each {} will be filled with the variables weight, height, width and word.

The speech to text will iterate through the audio file finding each word and retrieving the beginning and end times for the word. It also will find the punctuation and remove the values that are not necessary for this code.

The code then removes the formatting that exported from the speech to text and rewraps it in the CSS styling. This new string is added to the result.

Once the speech to text has finished running through the audio file, it will then finish the HTML file with </body></html> which then allows it to be processed as a completed file.

Now the options for the HTML to image are set to a BMP with page width of 384 pixels and the imgkit function is called and processes the result string.

This file is saved as message.bmp and the HTML file is saved and closed.

message.bmp is now sent to the thermal printer using the printer.printBitmap() function.

The device now will be able to react to the next time the toggle switch is flicked.

CONTRIBUTION

Throughout the project my tasks and contributions were many. At the commencement of semester I contributed to the conception of the idea and determination of potential avenues of project exploration.

After the focus of the project was determined I was responsible for researching what technologies were useful for our project, as well as highlighting what areas of theory should be investigated by the other group members. All members contributed to this investigative phase.

After the primary background research was completed, as a team we determined what areas each member was interested in learning so we could equitably distribute the workload based on expertise and learning goals. Because of my understanding of the technologies and the theories, I helped to guide each team member to ascertain the required avenues of exploration for completing their tasks. This also involved coaching members to develop skills, to ensure project timelines were met.

The initial tasks delegated to me were the research and development of the internal programming such as the speech

to text and connectivity of the various technologies, investigating which exact technologies were required and how they could be manipulated.

These tasks then expanded to consider how the audio is recorded and processed, how the typography is formed, how the modulation devices such as the EMF antenna work and how to print the receipt.

Following group bodystorming, the development of storyboards and face-to-face (where possible) and virtual simulations, I was then responsible for the final design and creation of the outer structure and possible manufacture.

In addition to these tasks I assisted team members with their designated activity through the provision of technical details and information.

For the presentation phases of our project throughout the semester I created sketches, renders, animations and other visual media including the compilation and production of the final video presentation, of which group members contributed individual content.

REFLECTION

Throughout the development of this project, I encountered many ideological and intellectual pathways of invaluable benefit. The semester gave me opportunity to develop skills in leadership as well as designing many aspects of a complex technology. It was obvious from the beginning that my aims and expectations were excessive in comparison to my peers. By setting my goals high I forced myself to increase my productivity and capabilities as both a group member and designer. The project required a vast array of technical ability such as programming in several languages, understanding obscure technologies and their interplay, and learning the complexities of modern sociolinguistic structures and cultures.

This self-enforced pressure was however, a double-edged sword. I may have had the ability to learn an enormous array of skills and theory, but it came at the cost of requiring a large amount of effort and time. This development also in ways seemed to alienate the team, as our learning trajectories and comprehension of the activities differed. This in turn led to much of my time leading, directing, and explaining the processes in intricate detail.

By the end of the semester, I was required to assist the other members with their delegated tasks to ensure the project progressed.

The project which at the beginning seemed like a very manageable task for four people became overwhelming at times. This paradigm was a consequence of my individual goals and a great learning experience, providing insight into how moving forward, I can both achieve the wanted level of self-benefit without detriment to the group dynamics.

I am thankful as I have gained a significant amount from the learning experience, especially in relation to leadership skills.

REFERENCES

- Argyle, M 1972, 'Non-verbal communication in human social interaction.',.
- Battle, S 2019, 'DRAWING PICTURES WITH A THEREMIN. Handsfree Etch-a-Sketch - YouTube', viewed 14 June 2021, <<https://www.youtube.com/watch?v=Rt3jlSQ7E1Y>>.
- Brown, TJ, Hooper, L, Elliott, RA, Payne, K, Webb, R, Roberts, C, Rostom, A & Symmons, D 2006, 'A comparison of the cost-effectiveness of five strategies for the prevention of non-steroidal anti-inflammatory drug-induced gastrointestinal toxicity: a systematic review with economic modelling.', *Health technology assessment* (Winchester, England), vol. 10, no. 38, pp. iii–iv, xi–xiii, 1–183.
- Bryant, GA & Fox Tree, JE 2005, 'Is there an Ironic Tone of Voice?', *Language and Speech*, vol. 48, no. 3, pp. 257–277.
- Castro, JCE, Martins, P, Boavida, A & MacHado, P 2019, '«máquina de Ouvir» - From Sound to Type: Finding the Visual Representation of Speech by Mapping Sound Features to Typographic Variables', in *ACM International Conference Proceeding Series*, Association for Computing Machinery.
- Cheang, HS & Pell, MD 2008, 'The sound of sarcasm', *Speech Communication*, vol. 50, no. 5, pp. 366–381.
- Ferrara, K, Brunner, H & Whittemore, G 1991, 'Interactive Written Discourse as an Emergent Register', *Written Communication*, vol. 8, no. 1, pp. 8–34.
- Goulart, L, Gray, B, Staples, S, Black, A, Shelton, A, Biber, D, Egbert, J & Wizner, S 2020, 'Linguistic Perspectives on Register', *Annual Review of Linguistics*, vol. 6, pp. 435–455.
- Hewes, G 1992, 'History of Glottogonic Theories', in *Language Origin: A Multidisciplinary Approach*, Springer Netherlands, pp. 3–20.
- Hunston, S 2007, 'Semantic prosody revisited', *International Journal of Corpus Linguistics*, vol. 12, no. 2, pp. 249–268.
- Iverson, JM & Goldin-Meadow, S 2005, 'Gesture paves the way for language development', *Psychological Science*, vol. 16, no. 5, pp. 367–371.
- James, A 2017, 'Prosody and paralanguage in speech and the social media: The vocal and graphic realisation of affective meaning', *Linguistica*, vol. 57, no. 1, pp. 137–149.
- Jensen, KE 2014, 'Linguistics in the digital humanities: (computational) corpus linguistics', *MedieKultur: Journal of media and communication research*, vol. 30, no. 57, p. 20.
- Key, MR 2011, *The relationship of verbal and nonverbal communication*, De Gruyter.
- Keyes, E 1993, 'Typography, color, and information structure', *Technical Communication*, vol. 40, no. 4, pp. 638–654.
- Klinke, CL 1986, 'Gaze and Eye Contact. A Research Review', *Psychological*

Bulletin, vol. 100, no. 1, pp. 78–100.

Luangrath, AW, Peck, J & Barger, VA 2017, 'Textual paralinguistic and its implications for marketing communications', *Journal of Consumer Psychology*, vol. 27, no. 1, pp. 98–107.

Marmpena, M, Lim, A & Dahl, TS 2018, 'How does the robot feel? Perception of valence and arousal in emotional body language', *Paladyn*, vol. 9, no. 1, pp. 168–182.

Mehndiratta, P, Sachdeva, S & Soni, D 2017, 'Detection of sarcasm in text data using deep convolutional neural networks', *Scalable Computing*, vol. 18, no. 3, pp. 219–228.

Nord Projects 'Little Printers', viewed 14 June 2021, <<https://nordprojects.co/projects/littleprinters/>>.

Pennycook, A 1985, 'Actions Speak Louder Than Words: Paralinguistic, Communication, and Education', *TESOL Quarterly*, vol. 19, no. 2, p. 259.

Prazdnichov, D 2018, 'Pocketsphinx Python', PyPi, viewed 13 June 2021, <<https://pypi.org/project/pocketsphinx/>>.

Rheindorf, M & Wodak, R 2019, 'Genre-related language change: Discourse- and corpus-linguistic perspectives on Austrian German 1970-2010', *Folia Linguistica*, vol. 53, no. 1, pp. 125–167.

Rosenberger, T & MacNeil, RL 1999, 'Prosodic font', in p. 252.

Shany-Ur, T, Poorzand, P, Grossman, SN, Growdon, ME, Jang, JY, Ketelle, RS, Miller, BL & Rankin, KP 2012, 'Comprehension of insincere communication in neurodegenerative disease: Lies, sarcasm, and theory of mind', *Cortex*, vol. 48, no. 10, pp. 1329–1341.

Spitzmüller, J 2019, *A Sociolinguistics of Typography*, St. Petersburg, viewed 10 June 2021, <www.spitzmueller.org/PetertheGreatSt.PetersburgPolytechnicUniversity>.

Toolan, M 2007, 'Trust and text, text as trust', *International Journal of Corpus Linguistics*, vol. 12, no. 2, pp. 269–288.

Tsonos, D & Kouroupetroglou, G 2008, 'A Methodology for the Extraction of Reader's Emotional State Triggered from Text Typography', in *Tools in Artificial Intelligence*, InTech.

Vgalin 'html2image', PyPi, viewed 10 June 2021, <<https://pypi.org/project/html2image/>>.

Walther, JB & Tidwell, LC 1995, 'Nonverbal cues in computer mediated communication, and the effect of chronemics on relational communication', *Journal of Organizational Computing*, vol. 5, no. 4, pp. 355–378.

Wang, Y, Shen, Y, Liu, Z, Liang, PP, Zadeh, A & Morency, LP 2019, 'Words can shift: Dynamically adjusting word representations using nonverbal behaviors', in *33rd AAAI Conference on Artificial Intelligence, AAAI 2019, 31st Innovative Applications of Artificial Intelligence Conference, IAAI 2019 and the 9th AAAI Symposium on Educational Advances in Artificial Intelligence, EAAI 2019*, AAAI Press, pp. 7216–7223.

